

QUESTION 24

Develop a Simple AI Assistant Using External Documents

External Knowledge Source: external_document.txt

Vector Database: FAISS | Embeddings: all-MiniLM-L6-v2 | LLM: FLAN-T5

1. Aim

To develop a simple AI assistant capable of answering user questions by retrieving relevant information from an external document and generating a concise response using a language model.

2. Problem Statement

A simple AI assistant should be able to use information supplied by an external document rather than depending only on built-in model knowledge. This implementation loads an external text file, converts its content into searchable embeddings, retrieves relevant sections, and uses the retrieved context to generate answers.

3. Objectives

- Load an external document into the application.
- Split the document into searchable chunks.
- Generate semantic embeddings for the chunks.
- Store embeddings in a FAISS vector database.
- Retrieve the Top-3 relevant document sections.
- Provide retrieved context to an AI assistant model.
- Generate answers to user questions from the external document.

4. System Architecture

External-Document AI Assistant Architecture

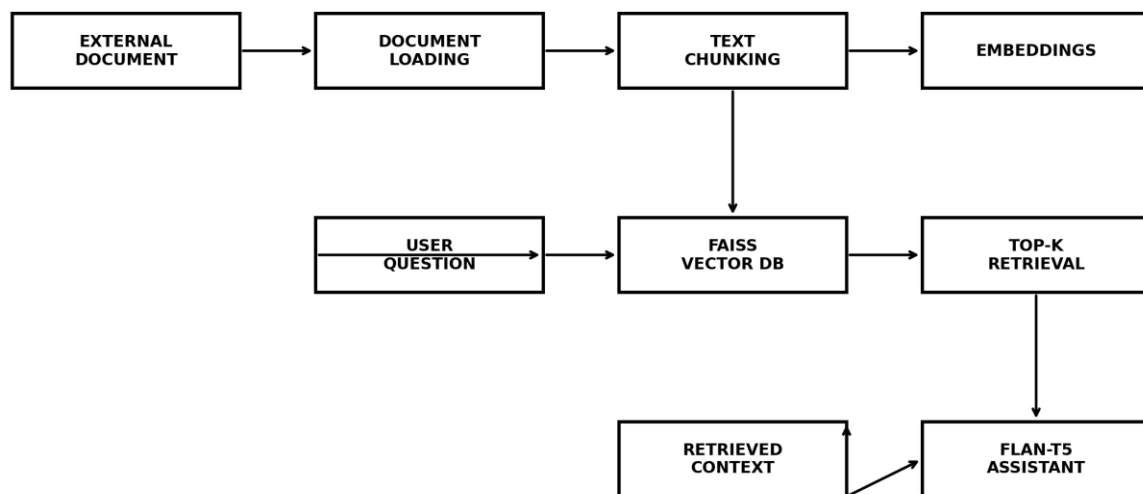


Figure 1: External-document AI assistant architecture.

5. Working Procedure

1. Load `external_document.txt`.
2. Split the document into text chunks.
3. Generate 384-dimensional embeddings using `all-MiniLM-L6-v2`.
4. Normalize and store embeddings in FAISS `IndexFlatIP`.
5. Convert each user question into an embedding.
6. Retrieve the three most relevant chunks.
7. Build context from the retrieved chunks.
8. Generate the answer using `google/flan-t5-small`.
9. Display retrieved chunk IDs, similarity scores, and the assistant response.

6. Technology Stack

Component	Implementation
Programming Language	Python
External Document	<code>external_document.txt</code>
Embedding Model	<code>all-MiniLM-L6-v2</code>
Embedding Dimension	384
Vector Database	FAISS
Index Type	<code>IndexFlatIP</code>
Top-K Retrieval	3
Generation Model	<code>google/flan-t5-small</code>

7. Python Implementation

```
with open("external_document.txt", "r", encoding="utf-8") as file:  
    text = file.read()
```

```
# Create text chunks
```

```
paragraphs = [p.strip() for p in text.split("\n\n") if p.strip()]  
chunks = []
```

```
for paragraph in paragraphs:
```

```
    words = paragraph.split()
```

```
    for i in range(0, len(words), 45):
```

```
        chunk = " ".join(words[i:i + 45])
```

```
        if chunk:
```

```
            chunks.append(chunk)
```

```
# Generate embeddings
```

```
embeddings = embedding_model.encode(  
    chunks, convert_to_numpy=True
```

```
).astype("float32")
```

```
faiss.normalize_L2(embeddings)
```

```

# Build vector database
index = faiss.IndexFlatIP(embeddings.shape[1])
index.add(embeddings)

# Retrieve relevant document sections
query_embedding = embedding_model.encode(
    [question], convert_to_numpy=True
).astype("float32")

faiss.normalize_L2(query_embedding)

scores, indices = index.search(
    query_embedding, 3
)

# Build context and generate answer
context = "\n\n".join(
    chunks[int(i)] for i in indices[0]
)

```

8. Test Questions

- What is artificial intelligence?
- What is supervised learning?
- What is deep learning?

9. Actual Implementation Output

```

DOCUMENT-BASED AI ASSISTANT

You: What is artificial intelligence?
Retrieved:
1. Chunk 0 (Score: 0.8613)
2. Chunk 1 (Score: 0.6230)
3. Chunk 5 (Score: 0.4425)
AI Assistant:
a field of computer science that focuses on creating systems capable
of performing tasks that normally require human intelligence

You: What is supervised learning?
Retrieved:
1. Chunk 2 (Score: 0.7696)
2. Chunk 1 (Score: 0.5508)
3. Chunk 3 (Score: 0.5155)
AI Assistant:
Unsupervised learning works with data without predefined labels.
It can discover hidden structures and patterns.

You: What is deep learning?
Retrieved:
1. Chunk 4 (Score: 0.8565)
2. Chunk 1 (Score: 0.5400)
3. Chunk 0 (Score: 0.4142)
AI Assistant:
specialized form of machine learning that uses multi-layer neural networks

```

Figure 2: Actual external-document AI assistant session.

10. Retrieval Results

Question	Top Chunk	Top Score	Assistant Response
What is artificial intelligence?	Chunk 0	0.8613	a field of computer science that focuses on creating systems capable of performing tasks that normally require human intelligence
What is supervised learning?	Chunk 2	0.7696	Unsupervised learning works with data without predefined labels. It can discover hidden structures and patterns.
What is deep learning?	Chunk 4	0.8565	specialized form of machine learning that uses multi-layer neural networks

11. Retrieval Score Analysis

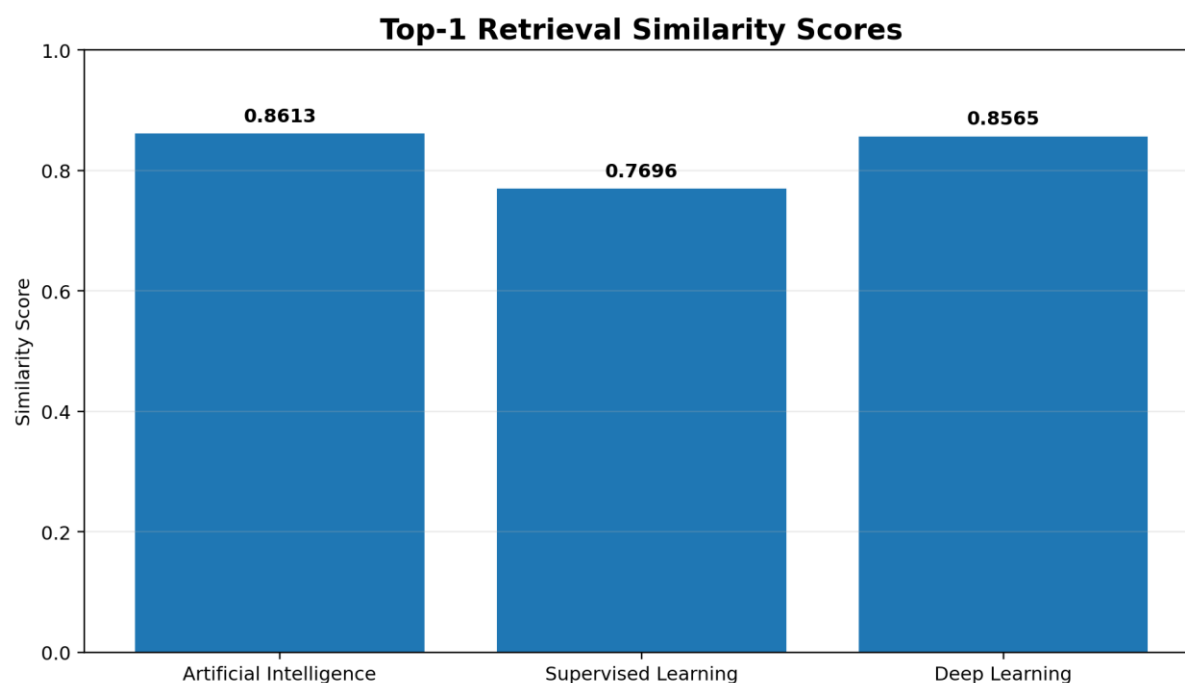


Figure 3: Highest similarity score for each tested question.

The highest Top-1 similarity score was 0.8613 for the Artificial Intelligence query. Deep Learning achieved 0.8565, while Supervised Learning achieved 0.7696. These scores indicate that the vector retrieval stage generally identified a relevant document section for each question.

12. Response Analysis

The Artificial Intelligence and Deep Learning responses were relevant to their questions. For the Supervised Learning question, the retriever correctly ranked Chunk 2 first with a score of 0.7696, but the generated response incorrectly described Unsupervised Learning. This demonstrates that retrieval can identify the correct source while a small language model may still generate an incorrect answer from the combined context.

13. Result

The external-document AI assistant successfully loaded `external_document.txt`, generated semantic embeddings, stored them in FAISS, retrieved Top-3 document sections, and generated answers for three questions. The implementation demonstrates a practical document-grounded AI assistant workflow.

14. Advantages

- Uses an external knowledge source.
- Semantic retrieval finds relevant document sections.
- Easy to update by replacing the external document.
- Provides source-grounded context to the language model.
- Can be extended to PDFs, Word files, and multiple documents.

15. Limitations Observed

- The small FLAN-T5 model can produce an incorrect answer even when the correct chunk is retrieved.
- Top-3 retrieval may include unrelated supporting chunks.
- The current implementation uses a text document as the external source.

16. Applications

- Document-based AI assistants
- Educational assistants
- Technical documentation assistants
- Knowledge-base search
- Enterprise document question answering

17. Conclusion

A simple AI assistant capable of answering questions using an external document was successfully implemented. The system loads `external_document.txt`, creates semantic embeddings, stores them in FAISS, retrieves relevant sections, and uses FLAN-T5 to generate responses. The highest observed retrieval score was 0.8613. The experiment also demonstrates an important practical limitation: correct retrieval does not always guarantee a correct generated answer when using a small language model.